

Graphs and Digraphs

Chapter 16

Chapter Contents

16.1 Directed Graphs

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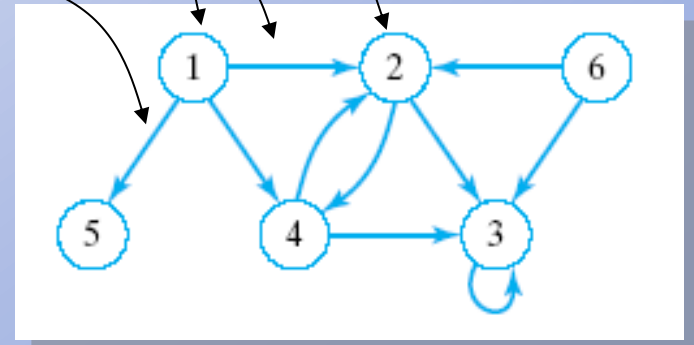
16.3 Graphs

Chapter Objectives

- Introduce directed graphs (digraphs) and look at some of the common implementations of them
- Study some of the algorithms for searching and traversing digraphs
- See how searching is basic to traversals and shortest path problems in digraphs
- Introduce undirected graphs and some of their implementations

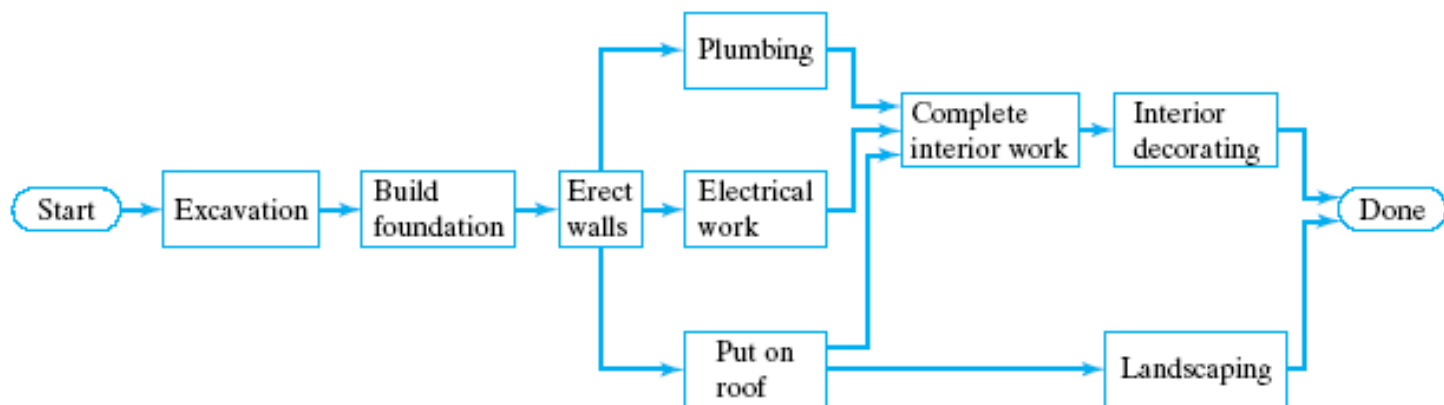
Directed Graphs

- Similar to a tree
- Consists of a finite set of elements
 - Vertices or nodes
- Together with finite set of directed
 - Arcs or edges
 - Connect pairs of vertices



Directed Graphs

- Applications of directed graphs
 - Analyze electrical circuits
 - Find shortest routes
 - Develop project schedules



Directed Graphs

- Trees are special kinds of directed graphs
 - One of their nodes (the root) has no incoming arc
 - Every other node can be reached from the node by a unique path
- Graphs differ from trees as ADTs
 - Insertion of a node does not require a link (arc) to other nodes ... or may have multiple arcs

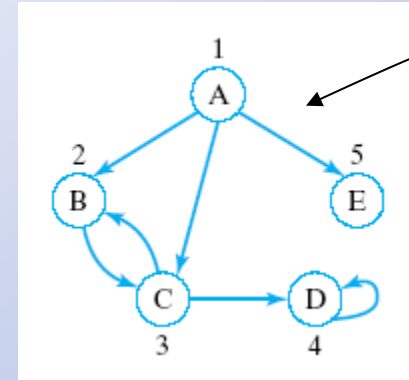
Directed Graphs

- A directed graph is defined as a collection of data elements:
 - Called nodes or vertices
 - And a finite set of direct arcs or edges
 - The edges connect pairs of nodes
- Operations include
 - Constructors
 - Inserts of nodes, of edges
 - Deletions of nodes, edges
 - Search for a value in a node, starting from a given node

Graph Representation

- Adjacency matrix representation
 - for directed graph with vertices numbered $1, 2, \dots, n$
- Defined as n by n matrix named adj
- The $[i, j]$ entry set to
 - 1 (true) if vertex j is adjacent to vertex i
(there is a directed arc from i to j)
 - 0 (false) otherwise

Graph Representation



columns j

	1	2	3	4	5
1	0	1	1	0	1
2	0	0	1	0	0
3	0	0	0	1	0
4	0	0	1	0	0
5	0	0	0	0	0

rows i

- Entry [1, 5] set to true
- Edge from vertex 1 to vertex 5

Graph Representation

- Weighted digraph
 - There exists a "cost" or "weight" associated with each arc
 - Then that cost is entered in the adjacency matrix
- A complete graph
 - has an edge between each pair of vertices
 - N nodes will mean $N * (N - 1)$ edges

Adjacency Matrix

- Out-degree of i^{th} vertex (node)
 - Sum of 1's (true's) in row i
- In-degree of j^{th} vertex (node)
 - Sum of the 1's (true's) in column j
- What is the out-degree of node 4?
- What is the in-degree of node 3?

Adjacency Matrix

- Consider the sum of the products of the pairs of elements from row i and column j

adj

	1	2	3	4	5
1	0	1	1	0	1
2	0	0	1	0	0
3	0	0	0	1	0
4	0	0	1	0	0
5	0	0	0	0	0

*adj*²

	1	2	3	4	5
1			1		
2					
3					
4					
5					

Fill in the rest
of *adj*²

This is the number of
paths of length 2 from
node 1 to node 3

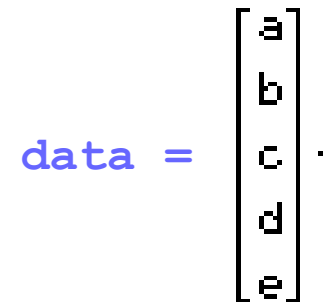
Adjacency Matrix

- Basically we are doing matrix multiplication
 - What is adj^3 ?
- The value in each entry would represent
 - The number of paths of length 3
 - From node i to node j
- Consider the meaning of the generalization of adj^n

Adjacency Matrix

- Deficiencies in adjacency matrix representation

- Data must be stored in separate matrix

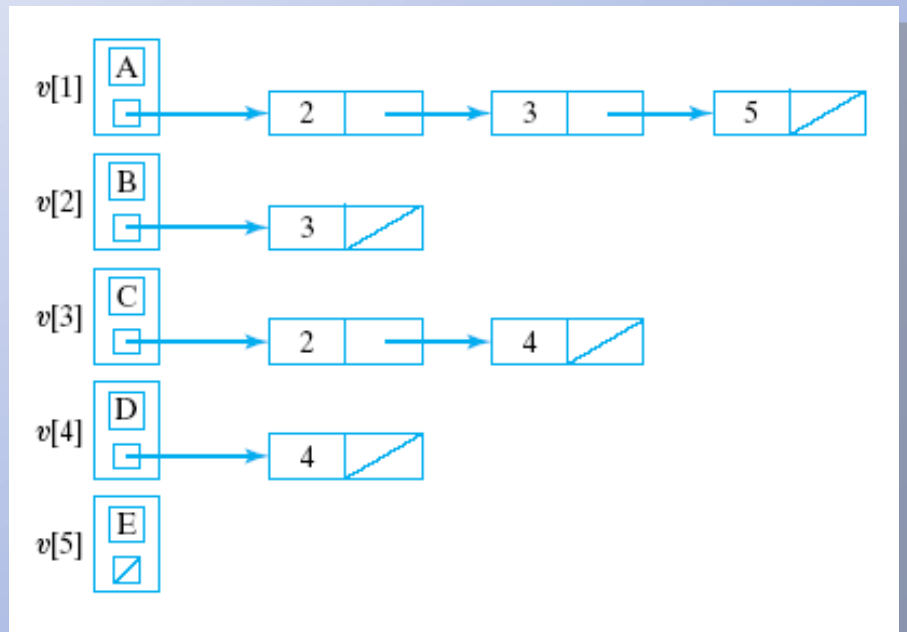


`data =` $\begin{bmatrix} a \\ b \\ c \\ d \\ e \end{bmatrix}.$

- When there are few edges the matrix is sparse (wasted space)

Adjacency-List Representation

- Solving problem of wasted space
 - Better to use an array of pointers to linked row-lists
- This is called an Adjacency-list representation
- View [source code](#) for class template



Searching a Graph

- Recall that with a tree we search from the root
- But with a digraph ...
 - may not be a vertex from which every other vertex can be reached
 - may not be possible to traverse entire digraph (regardless of starting vertex)

Searching a Graph

- We must determine which nodes are reachable from a given node
- Two standard methods of searching:
 - Depth first search
 - Breadth first search

Depth-First Search

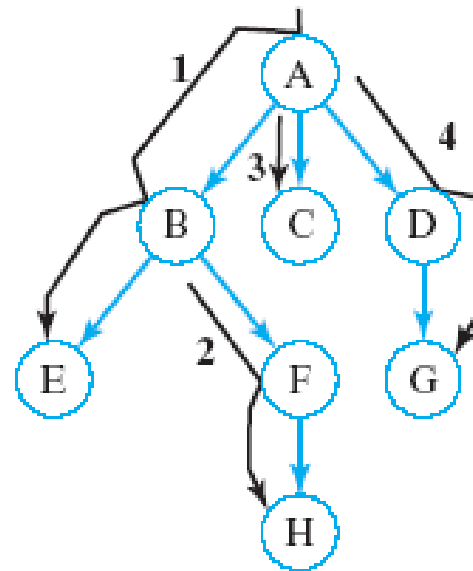
- Start from a given vertex v
- Visit first neighbor w , of v
- Then visit first neighbor of w which has not already been visited
- etc. ... Continues until
 - all nodes of graph have been examined
- If dead-end reached
 - backup to last visited node
 - examine remaining neighbors

Depth-First Search

- Start from node 1
- What is a sequence of nodes which would be visited in DFS?

Click for answer

A, B, E, F, H, C, D, G



Depth-First Search

- DFS uses backtracking when necessary to return to some values that were
 - already processed or
 - skipped over on an earlier pass
- When tracking this with a stack
 - pop returned item from the stack
- Recursion is also a natural technique for this task
- Note: DFS of a tree would be equivalent to a preorder traversal

Depth-First Search

Algorithm to perform DFS search of digraph

1. Visit the start vertex, v
2. For each vertex w adjacent to v do:
 If w has not been visited,
 apply the depth-first search algorithm
 with w as the start vertex.

Note the recursion

Breadth-First Search

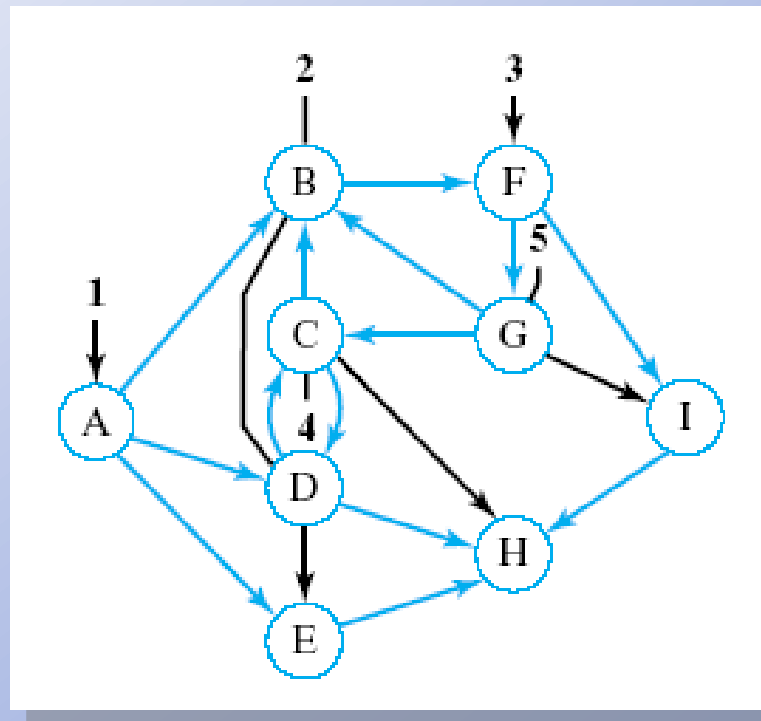
- Start from a given vertex v
- Visit all neighbors of v
- Then visit all neighbors of first neighbor w of v
- Then visit all neighbors of second neighbor x of v ... etc.
- BFS visits nodes by level

Breadth-First Search

- Start from node containing A
- What is a sequence of nodes which would be visited in BFS?

Click for answer

A, B, D, E, F, C, H, G, I



Breadth-First Search

- While visiting each node on a given level
 - store it so that
 - we can return to it after completing this level
 - so that nodes adjacent to it can be visited
- First node visited on given level should be First node to which we return

What data structure does this imply?

A queue

Breadth-First Search

Algorithm for BFS search of a diagraph

1. Visit the start vertex
 2. Initialize queue to contain only the start vertex
 3. While queue not empty do
 - a. Remove a vertex v from the queue
 - b. For all vertices w adjacent to v do:
If w has not been visited then:
 - i. Visit w
 - ii. Add w to queue
- End while

Graph Traversal

Algorithm to traverse digraph must:

- visit each vertex exactly once
- BFS or DFS forms basis of traversal
- Mark vertices when they have been visited

1. Initialize an array (vector) **unvisited**
unvisited[i] false for each vertex **i**
2. While some element of **unvisited** is false
 - a. Select an unvisited vertex **v**
 - b. Use BFS or DFS to visit all vertices reachable from **v**

End while

Paths

- Routing problems – find an optimal path in a network
 - a shortest path in a graph/digraph
 - a cheapest path in a weighted graph/digraph
- Example – a directed graph that models an airline network
 - vertices represent cities
 - direct arcs represent flights connecting cities
- Task: find most direct route (least flights)

Paths

- Most direct route equivalent to
 - finding length of shortest path
 - finding minimum number of arcs from start vertex to destination vertex
- Search algorithm for this shortest path
 - an easy modification of the breadth-first search algorithm

Shortest Path Algorithm

1. Visit **start** and label it with a **0**
 2. Initialize **distance** to **0**
 3. Initialize a queue to contain only **start**
 4. While **destination** not visited and the queue not empty do:
 - a. Remove a vertex **v** from the queue
 - b. If label of **v** > **distance**, set **distance++**
 - c. For each vertex **w** adjacent to **v**
If **w** has not been visited then
 - i. Visit **w** and label it with **distance + 1**
 - ii. Add **w** to the queueEnd for
- End while

Shortest Path Algorithm

5. If **destination** has not been visited then display
"Destination not reachable from start vertex"

else

Find vertices **p[0] ... p[distance]** on shortest path as follows

a. Initialize **p[distance]** to **destination**

b. For each value of **k** ranging from
distance - 1 down to **0**

Find a vertex **p[k]** adjacent to **p[k+1]** with label **k**

End for

-
- Note source code of Digraph Class Template, [Fig. 16.1](#)
 - View program to find shortest paths in a network, [Fig. 16.2](#)

NP-Complete Problems

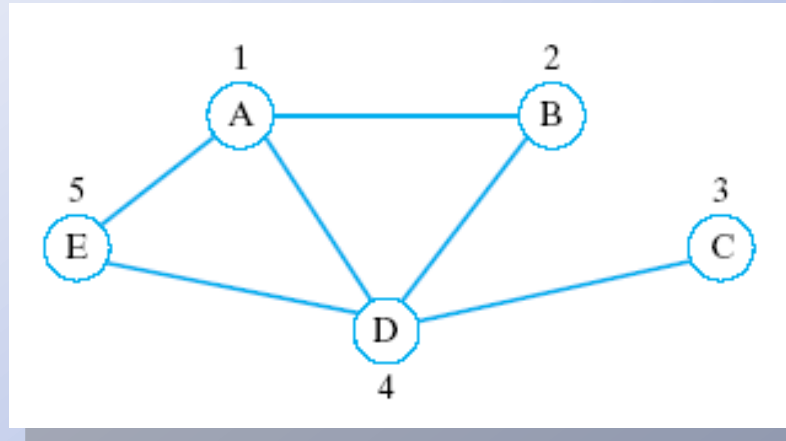
- Nondeterministic polynomial problems
 - Problems for which a solution can be guessed, then checked with an algorithm
 - Algorithm has computing time $O(P(n))$ for some polynomial $P(n)$
- Contrast deterministic polynomial (or P) problems
 - Can be solved by algorithms in polynomial time

NP-Complete Problems

- These are applied to shortest path problems
 - Example is traveling salesman problem
 - Find route to all destinations with least cost
- NP-Complete problems
 - If a polynomial time algorithm that solves any one of these problems can be found
 - Then the existence of polynomial time algorithms for all NP problems is guaranteed

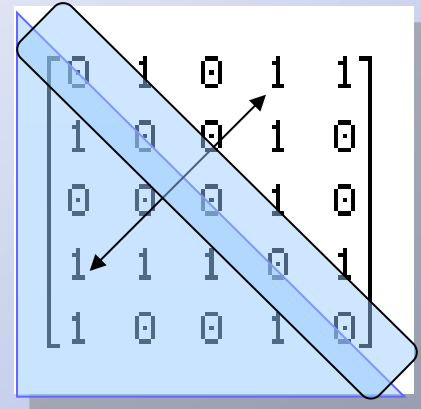
Graphs

- Like a digraph
 - Except no direction is associated with the edges
 - No edges joining a vertex to itself allowed



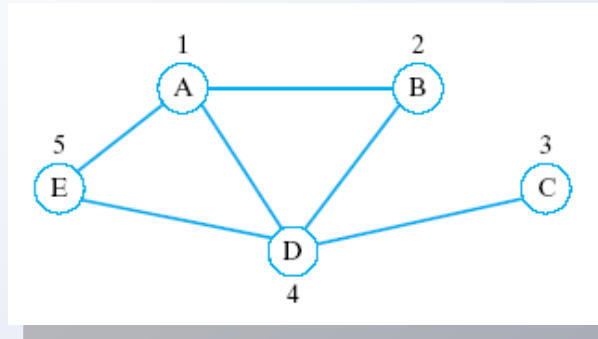
Undirected Graph Representation

- Can be represented by
 - Adjacency matrices
- Adjacency matrix will always be symmetric
 - For an edge from i to j , there must be
 - An edge from j to i
 - Hence the entries on one side of the matrix diagonal are redundant
- Since no loops,
 - the diagonal will be all 0's

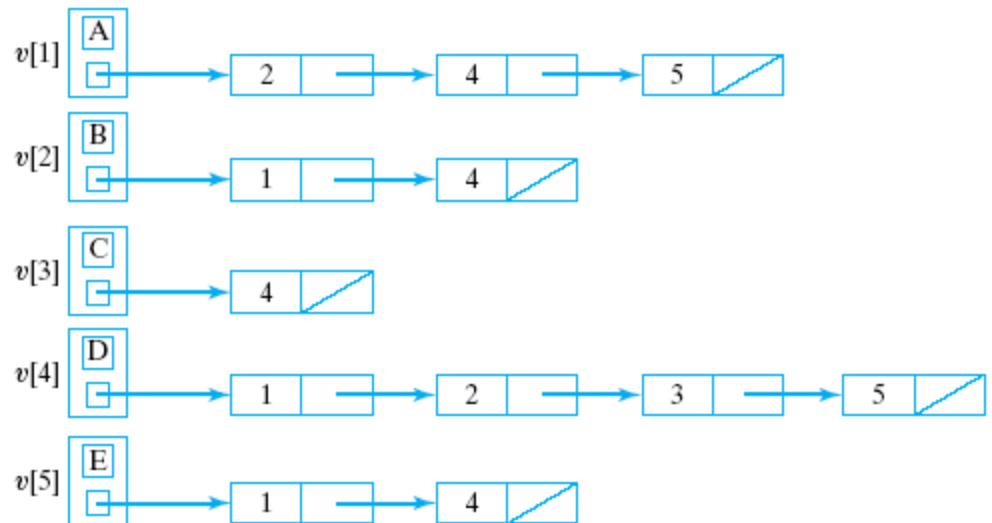


Undirected Graph Representation

- Given



- Adjacency-List representation

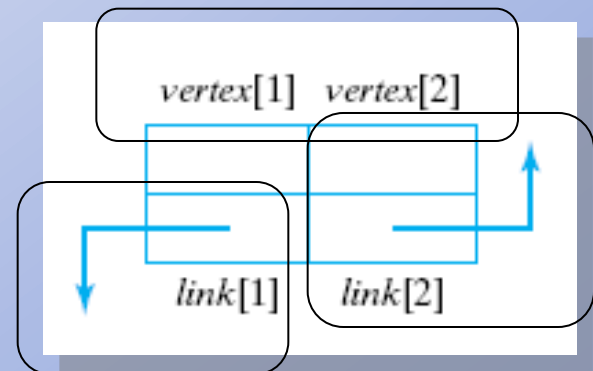


Edge Lists

- Adjacency lists suffer from the same redundancy
 - undirected edge is repeated twice
- More efficient solution
 - use edge lists
- Consists of a linkage of edge nodes
 - one for each edge
 - to the two vertices that serve as the endpoints

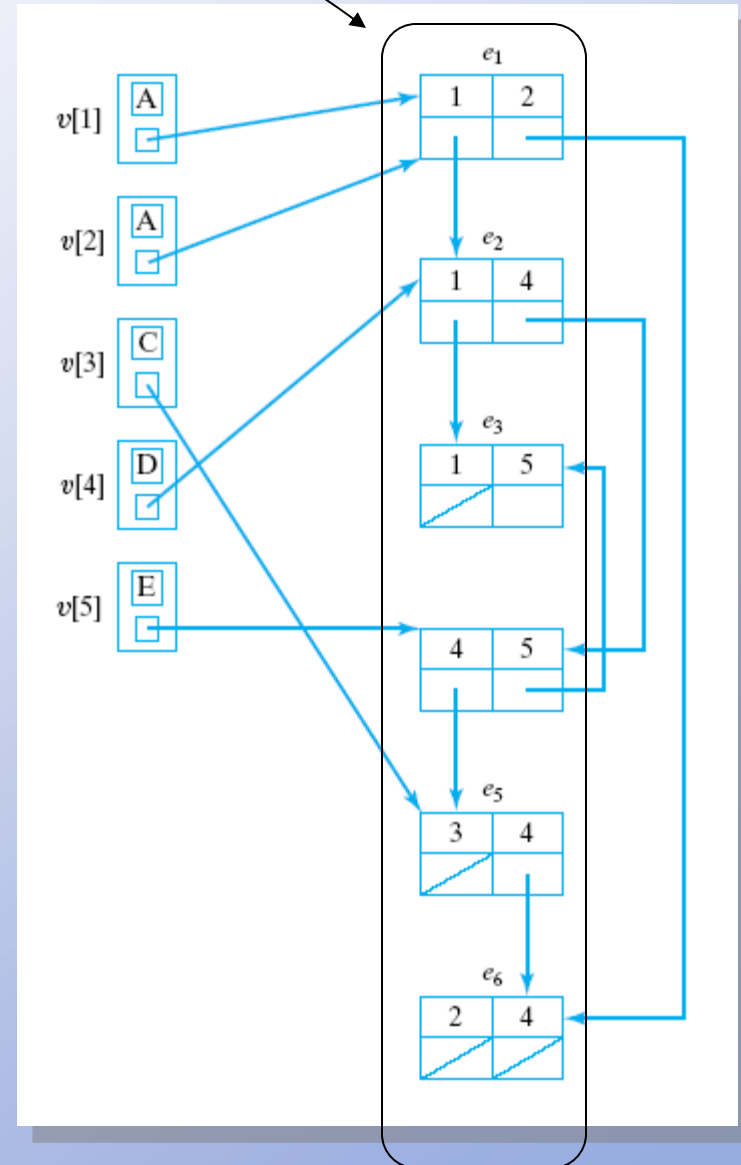
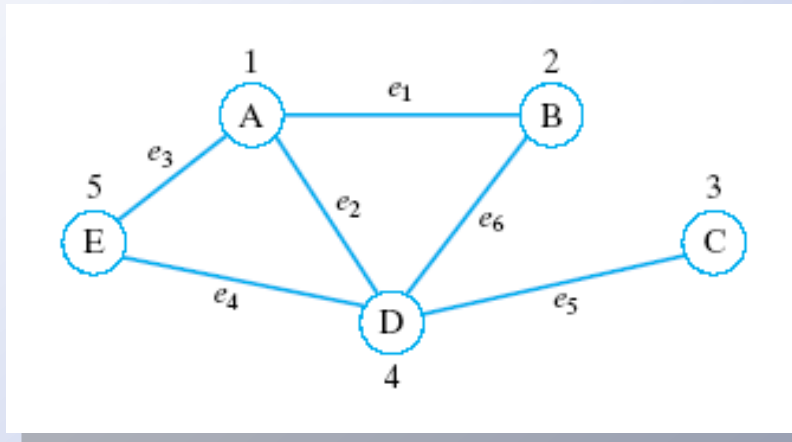
Edge Nodes

- Each edge node represents one edge
 - `vertex[1]` and `vertex[2]` are vertices connected by this edge
 - `link[1]` points to another edge node having `vertex[1]` as one end point
 - `link[2]` points to another edge node having `vertex[2]` as an endpoint



Edge List

- Vertices have pointers to one edge



Graph Operations

- DFS, BFS, traversal, etc. are similar as those for digraphs
- Note class template Graph, [Fig. 16.4](#)
 - Uses edge-list representation of graphs as just described

Connectedness

- Connected defined
 - A path exists from each vertex to every other vertex
- Note the `isConnected()` function in `Graph` class template
 - Uses a DFS, marks all vertices reachable from vertex 1
- View program in [Fig. 16-5](#)
 - Exercises this function